

A ()

A ()

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A ()

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A

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- Python
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A ():

A

(r, θ)
 $\vec{r}$

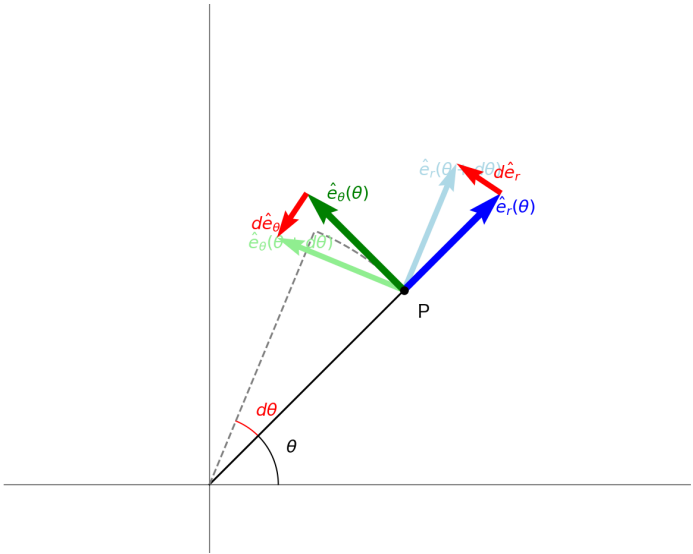
$\hat{e}_r$
 $\hat{e}_\theta$

$(\hat{i}, \hat{j})$
 $\hat{e}_r, \hat{e}_\theta$

$\theta \rightarrow \theta + d\theta$
 $d\hat{e}_r, d\hat{e}_\theta$

$\vec{r} = r\hat{e}_r$

極座標の単位ベクトルの微分（定性的な図解）



1:

$$\begin{array}{ccccccc}
 & 1. \frac{d\hat{e}_r}{d\hat{e}_\theta} : \hat{e}_r & \frac{d\theta}{\hat{e}_r} & d\hat{e}_r & \hat{e}_\theta & 1 \cdot d\theta & 2. \frac{d\hat{e}_\theta}{d\hat{e}_r} : \hat{e}_\theta \\
 d\theta & & & & & & \\
 dt & & & & & & \\
 & & \frac{d\hat{e}_r}{dt} = \dot{\theta}\hat{e}_\theta, & \frac{d\hat{e}_\theta}{dt} = -\dot{\theta}\hat{e}_r & & &
 \end{array}$$

quarto-live Pyodide
NumPy + Matplotlib

Python
Jupyter Notebook

WebAssembly Pyodide

```
///  
viewof length      = Inputs.range([0.1, 2.0], {step: 0.1, label: tex`\text{ } L \text{ }  
viewof initial_angle = Inputs.range([-1.57, 1.57], {step: 0.1, label: tex`\text{ } \theta_  
viewof gravity      = Inputs.range([1.0, 20.0], {step: 0.1, label: tex`\text{ } g \text{ }  
viewof time         = Inputs.range([0.0, 10.0], {step: 0.1, label: tex`\text{ } t \text{ }  
viewof animate      = Inputs.toggle({label: " ", value: false})  
  
//  
update_time = {  
  if (!animate) return;  
  while (animate) {  
    let next_t = time + 0.1;  
    if (next_t > 10.0) next_t = 0.0;  
    viewof time.value = next_t;  
    viewof time.dispatchEvent(new CustomEvent("input"));  
    await Promises.delay(100);  
  }  
}
```

```
#| autorun: true  
#| input: ["length", "initial_angle", "gravity", "time"]  
# --- (Noto Sans JP) - ---  
import os  
import warnings  
import matplotlib  
from matplotlib import font_manager
```

```

from pyodide.http import pyfetch

warnings.simplefilter("ignore") # font      Glyph missing

if not os.path.exists("/tmp/NotoSansJP.ttf"):
    resp = await pyfetch("../.../assets/fonts/NotoSansJP-Regular.ttf")
    with open("/tmp/NotoSansJP.ttf", "wb") as f:
        f.write(await resp.bytes())
    font_manager.fontManager.addfont("/tmp/NotoSansJP.ttf")
    matplotlib.rcParams["font.family"] = "Noto Sans JP"
    matplotlib.rcParams["axes.unicode_minus"] = False

# ---      ---
import numpy as np
import matplotlib.pyplot as plt

#      ( )
omega = np.sqrt(gravity / length) #
theta = initial_angle * np.cos(omega * time) #      (t)

#
x = length * np.sin(theta)
y = -length * np.cos(theta) #

fig, ax = plt.subplots(figsize=(6, 6))
ax.set_xlim(-2.5, 2.5)
ax.set_ylim(-2.5, 0.5)
ax.set_aspect("equal")
ax.grid(True)
ax.set_title(f"      t = {time:.1f} s      ")

ax.plot(0, 0, "ko", markersize=8) #
ax.plot([0, x], [0, y], "k-", lw=2) #
ax.plot(x, y, "ro", markersize=20) #

plt.tight_layout()
plt.show()

```

l m

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l} \sin \theta$$

$$\theta \ll 1 \qquad \sin \theta \approx \theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l} \theta$$

$$\theta(t) = \theta_0 \cos \left(\sqrt{\frac{g}{l}} t + \phi \right)$$

$$r = ae^{k\theta} \ (\theta > 0) \quad \dot{\theta} = \omega = \text{const} \quad v_r, v_\theta, a_r, a_\theta$$

$$a > 0 \quad a_r \propto k$$

k

```

//| echo: false
viewof k      = Inputs.range([-0.5, 0.5], {step: 0.05, label: tex`\text{ } k`, value: 0.15}
viewof omega = Inputs.range([0.5, 5.0], {step: 0.1, label: tex`\text{ } \omega`, value: 0.5}
viewof t      = Inputs.range([0.0, 5.0], {step: 0.05, label: tex`\text{ } t \text{ (s)}`, value: 0.0}
viewof animate = Inputs.toggle({label: "animate", value: false})

//
update_t = {
  if (!animate) return;
  while (animate) {
    let next_t = t + 0.05;
    if (next_t > 5.0) next_t = 0.0;
    viewof t.value = next_t;
    viewof t.dispatchEvent(new CustomEvent("input"));
    await Promises.delay(50);
  }
}

```

```

#| autorun: true
#| input: ["k", "omega", "t"]
# ---      (Noto Sans JP)      -      ---
import os
import warnings
import matplotlib
from matplotlib import font_manager
from pyodide.http import pyfetch

warnings.simplefilter("ignore") # font      Glyph missing

if not os.path.exists("/tmp/NotoSansJP.ttf"):
    resp = await pyfetch("../.../assets/fonts/NotoSansJP-Regular.ttf")
    with open("/tmp/NotoSansJP.ttf", "wb") as f:
        f.write(await resp.bytes())
    font_manager.fontManager.addfont("/tmp/NotoSansJP.ttf")
    matplotlib.rcParams["font.family"] = "Noto Sans JP"
    matplotlib.rcParams["axes.unicode_minus"] = False

# ---      ---
import numpy as np
import matplotlib.pyplot as plt

a = 1.0 #

#
theta_max = max(omega * t, 0.01) # 0
theta_array = np.linspace(0, theta_max, 200)

#      (r, ) →      (x, y)
r_array = a * np.exp(k * theta_array)
x_array = r_array * np.cos(theta_array)
y_array = r_array * np.sin(theta_array)

#
x_current, y_current = x_array[-1], y_array[-1]

#      t=5 s
max_r = max(a, a * np.exp(abs(k) * omega * 5.0))

fig, ax = plt.subplots(figsize=(6, 6))
ax.set_xlim(-max_r * 1.1, max_r * 1.1)

```

```

ax.set_ylim(-max_r * 1.1, max_r * 1.1)
ax.set_aspect('equal')
ax.grid(True, linestyle='--', alpha=0.7)
ax.set_title(f"      (t = {t:.1f} s, k = {k:.2f},    = {omega:.1f})")

ax.plot(x_array, y_array, 'k-', alpha=0.5, label=' ')
ax.plot([0, x_current], [0, y_current], 'b-', lw=1.5, label='      $\vec{r}$')
ax.plot(x_current, y_current, 'ro', markersize=8, label=' ')

ax.legend(loc='upper left')
plt.tight_layout()
plt.show()

```

1.

$$\vec{r} = r\hat{e}_r \quad t$$

$$(AB)' = A'B + AB'$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{e}_r) = \dot{r}\hat{e}_r + r\dot{\hat{e}}_r$$

$$\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta$$

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$v_r = \dot{r}, v_\theta = r\dot{\theta}$$

$$3 \quad (ABC)' = A'BC + AB'C + ABC'$$

$$\begin{aligned} \vec{a} &= \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{e}_r) + \frac{d}{dt}(r\dot{\theta}\hat{e}_\theta) \\ &= (\ddot{r}\hat{e}_r + \dot{r}\dot{\hat{e}}_r) + (\dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\dot{\hat{e}}_\theta) \end{aligned}$$

$$\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta \quad \dot{\hat{e}}_\theta = -\dot{\theta}\hat{e}_r$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

$$a_r, a_\theta$$

2.

$$\begin{aligned}
 * r &= ae^{k\theta} * \dot{\theta} = \omega \quad () * \ddot{\theta} = \frac{d}{dt}(\omega) = 0 \\
 r & \quad \dot{r} \quad \theta \quad t \quad r(t) = ae^{k\theta(t)} \quad t \quad \theta \quad \dot{\theta} \\
 \dot{r} &= \frac{dr}{d\theta} \cdot \frac{d\theta}{dt} = \frac{d}{d\theta}(ae^{k\theta}) \cdot \omega = (ake^{k\theta}) \cdot \omega = ak\omega e^{k\theta} \\
 ae^{k\theta} &= r \\
 \dot{r} &= k\omega r
 \end{aligned}$$

$$\begin{aligned}
 \ddot{r} &= \frac{d}{dt}(C \cdot f(t)) = C \cdot \dot{f}(t) \\
 \ddot{r} &= \frac{d}{dt}(k\omega r) = k\omega \cdot \dot{r} \\
 \dot{r} &= ak\omega e^{k\theta} \\
 \ddot{r} &= k\omega(ak\omega e^{k\theta}) = ak^2\omega^2 e^{k\theta}
 \end{aligned}$$

:

$$\begin{aligned}
 v_r &= \dot{r} = ak\omega e^{k\theta} \\
 v_\theta &= r\dot{\theta} = (ae^{k\theta})\omega = a\omega e^{k\theta}
 \end{aligned}$$

:

$$\begin{aligned}
 a_r &= \ddot{r} - r\dot{\theta}^2 = ak^2\omega^2 e^{k\theta} - (ae^{k\theta})\omega^2 = a\omega^2(k^2 - 1)e^{k\theta} \\
 a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} = (ae^{k\theta})(0) + 2(ak\omega e^{k\theta})\omega = 2ak\omega^2 e^{k\theta}
 \end{aligned}$$

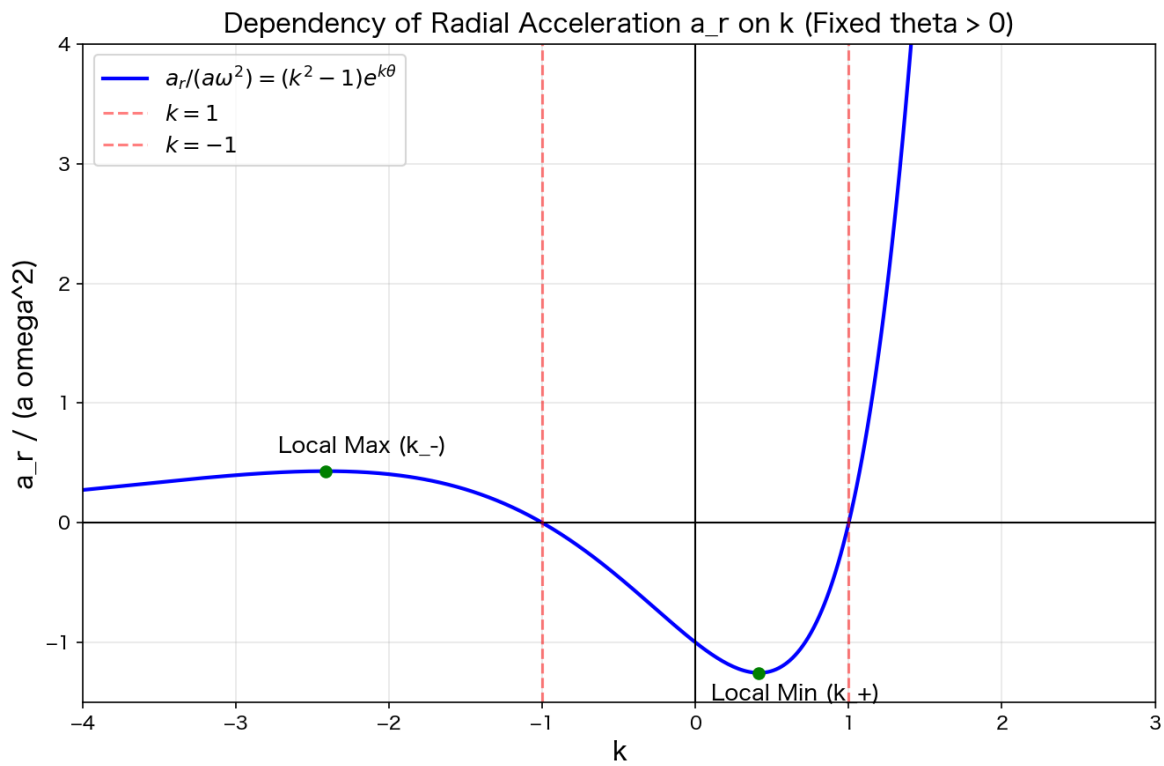
3. $a_r = k$

$$\begin{aligned}
 a_r(k) &= a\omega^2(k^2 - 1)e^{k\theta} \quad a > 0 \quad \theta > 0 \quad k \quad a\omega^2 > 0 \\
 f(k) &= (k^2 - 1)e^{k\theta}
 \end{aligned}$$

$$\begin{aligned}
 f'(k) &= 2ke^{k\theta} + (k^2 - 1) \cdot \theta e^{k\theta} = (\theta k^2 + 2k - \theta)e^{k\theta} \\
 f'(k) &= 0 \quad k = 2 \quad \theta k^2 + 2k - \theta = 0
 \end{aligned}$$

$$k_{\pm} = \frac{-1 \pm \sqrt{1 + \theta^2}}{\theta}$$

$$\theta > 0 \quad k_+ > 0 \quad k_- < -1$$



1: a_r k

$$1. \quad a_r : e^{k\theta} \quad a_r (k^2 - 1)$$

- $|k| < 1 : a_r < 0$
- $|k| > 1 : a_r > 0$
- $k = \pm 1 : a_r = 0$

$$2. \quad k \rightarrow \infty : e^{k\theta}$$

$$3. \quad k \rightarrow -\infty : (k^2 - 1) e^{k\theta} \rightarrow 0 \quad a_r \rightarrow 0$$

$$4. \quad : k < -1 \quad 0 < k < 1$$

0